$\Delta L = n \Delta x$

- (optical path) is greatly changed!
- Fringes move/shift.
- What about the total no. of fringes!?

→ surface roughness.
→ film thickness
→ nilnosopic displacements
→ vibrations (nm) scale.

→ form of measured surface.
↳ structure?

(length-measuring)

☆ changing mirror position → shift fringes → optical path diff

Light:
☆ → interferometer → with temporal coherence → produce observable fringes.

Fringe Visibility as a fn of mirror displacement = Autocorrelation fn. of light.

with Wiener-Khinchin theorem (Fourier transformation) we can reconstruct the spectral distribution of white light.

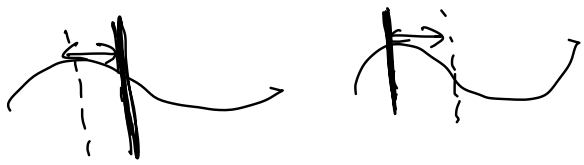
↳ what wavelengths are inside the WL.
↳ each wavelength intensity (contribution).

Optical Path Length diff (Δz), Partial waves, Coherently superimposed.

∴ (like constructive/destructive) interference

Geometrical Path Length diff ($\Delta \phi$)

Diff (~) ↗ length of arms.
($x_2 - x_1 = \Delta x$)



(Par. D)

$\Delta Z \longrightarrow$ time delay $\tau \longrightarrow$ radiant intensity $I(\tau, \vartheta)$

ϑ (light freq)